

Comparative Analysis: BTC vs S&P 500 — Return and Risk (2025).

The purpose of this portfolio is to clearly demonstrate whether these assets are suitable for diversification and whether they are worth the risk or investment using financial metrics.

Analysis tools: TradingView, SQL (NebulaTool), Data Visualization (Datawrapper), Google Sheets.

3 Key Facts:

1. 2025 was a very eventful year, marked by sharp declines and sudden surges.
2. Correlation: The trends of these diverse assets sometimes move in sync down to the daily level (for example, both assets fell from February 24 to on April 7, 2025, followed by a sharp rise).
3. A strategy for investment portfolios: Bitcoin is suitable for an aggressive portfolio (high risk/return), while the index is preferable for a conservative one (stability and predictability).

Data Sources:

Historical Prices (BTC & S&P 500): <https://tradersunion.com/>

Macro Data (Risk-Free Rate): <https://fred.stlouisfed.org/series/DGS10>,
<https://tradingeconomics.com/united-states/inflation-cpi>

1. Data preparation and filtering (SQL), calculation of daily/annual returns.

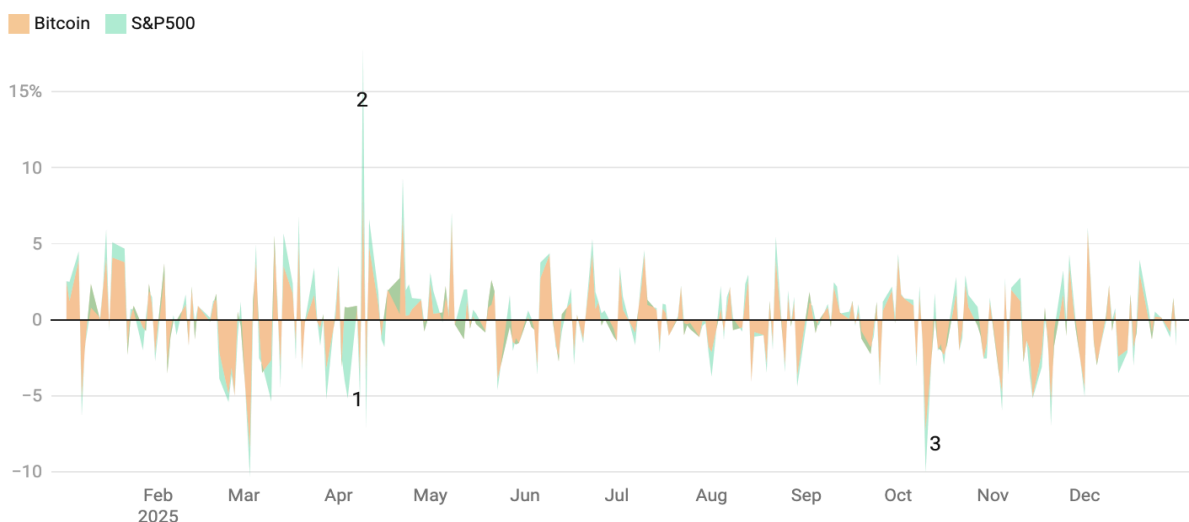
Used SQL to calculate the daily return in % + Google Sheets formulas for the rest.

```

WITH btc AS (
  SELECT date,
  close AS btc_close,
  ROUND (((close - LAG(close) OVER (ORDER BY date)) / LAG(close) OVER (ORDER BY date)) * 100, 2) AS
  btc_return_perc
  FROM btc2
),
sp AS (
  SELECT date,
  close AS sp_close,
  ROUND (((close - LAG(close) OVER (ORDER BY date)) / LAG(close) OVER (ORDER BY date)) * 100, 2) AS
  sp_return_perc
  FROM sp500_2
)
SELECT b.date,
b.btc_close,
s.sp_close,
b.btc_return_perc,
s.sp_return_perc
FROM btc AS b
INNER JOIN sp AS s ON b.date = s.date
ORDER BY b.date

```

Daily return



1. April 3–4: A sharp market decline following Trump’s imposition of new tariffs.
2. Sharp 8-10% rise on April 7–9—a reaction to the 90-day tariff pause.
3. Bitcoin loses a new all-time high, with over \$19 billion in liquidations and a market in overdrive.

2025 total return:

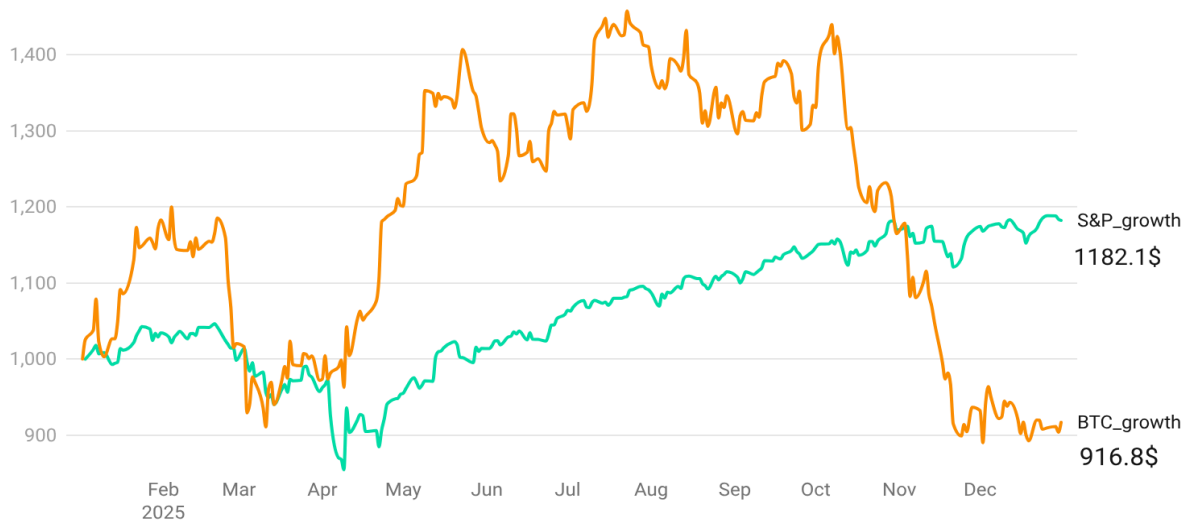
Bitcoin **-9.62%**

S&P500 **+16.64%** (+1.5% in dividends)

 **The main point:** “Despite reaching an all-time high, Bitcoin has has a negative year, in contrast to the stable and slowly rising S&P 500 benchmark.”

2. Simulator for investing \$1,000 in each asset + Sharpe Ratio.

Growth of 1,000\$ investment



Created with Datawrapper

(Click on the link in the image to view the chart in more detail).

Sharpe Ratio:

$$S_{BTC} = \frac{-0.0962 - 0.043}{0.37} = \frac{-0.1392}{0.37} \approx -0.38 \quad S_{SP500} = \frac{+0.1884 - 0.043}{0.18} = \frac{+0.1454}{0.18} \approx +0.81$$

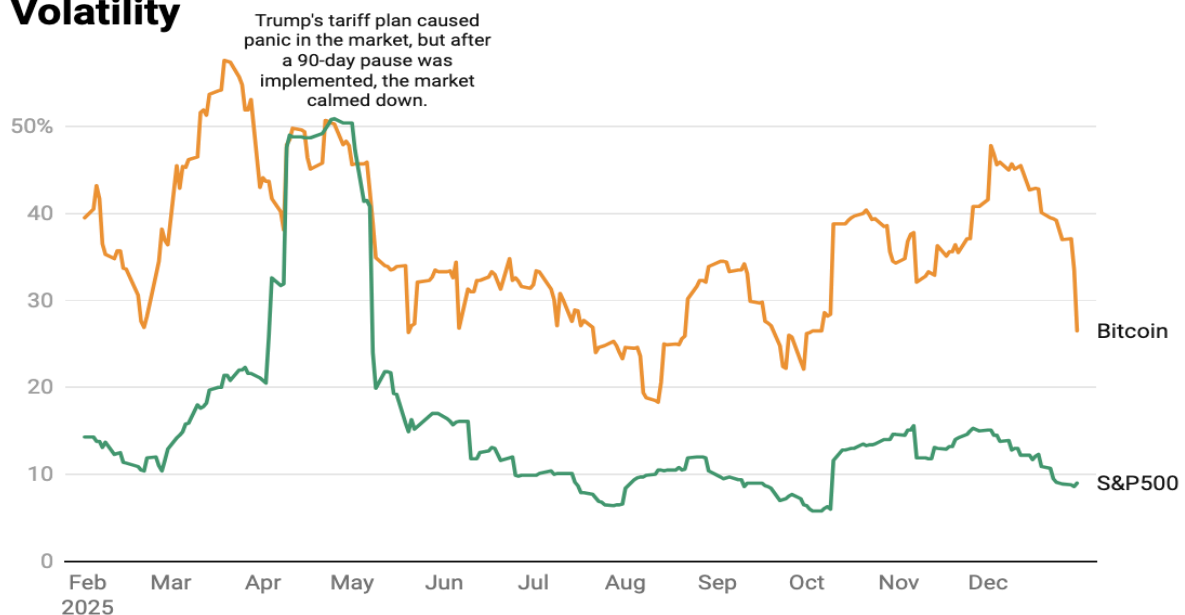
Bitcoin (Sharpe < 0): **A disaster in terms of performance.** A negative value means that the asset isn't just risky—it's unprofitable compared to a standard bank deposit or government bonds. The investor was "rewarded" with immense stress and negative return.

S&P 500 (Sharpe ≈ 0.8): **The benchmark for reliability.** A value close to 1 confirms that every risk taken has been generously rewarded by the market. This is not a game, but a professional tool for capital growth.

⚖️ The main point: A \$1,000 investment simulator combined with the Sharpe ratio clearly demonstrates that the hyped-up, modern Bitcoin—which is subject to manipulation by politicians and financial whales—falls significantly short of the classic, benchmark S&P 500 index. Until the fall, it was possible to make money on Bitcoin (while losing your nerves) when Trump became president and began actively promoting Bitcoin (we all know for what purposes), but there is one golden rule in financial markets: if an asset rises too quickly, it will fall just as quickly. Bitcoin has still not (as of June 6, 2026—the date of this analysis) been able to recover and continues to fall.

3. Volatility+Maxdrawdown.

Volatility



(Click on the link in the image to view the chart in more detail).

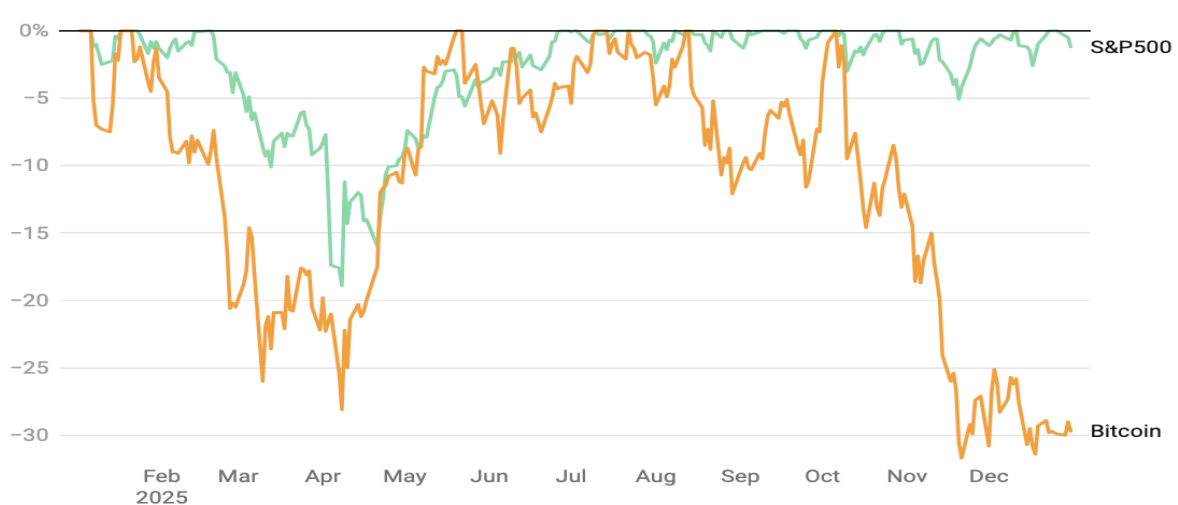
Annual volatility:

Bitcoin - 37%

S&P - 18%

Key takeaway: The chart shows that in April 2025, the S&P 500 exhibited extreme sensitivity to macroeconomic shocks (Trump's tariff imposition), temporarily surpassing even Bitcoin in terms of volatility. This suggests that during periods of global uncertainty, traditional stock markets can become riskier than cryptocurrency.

Drawdown

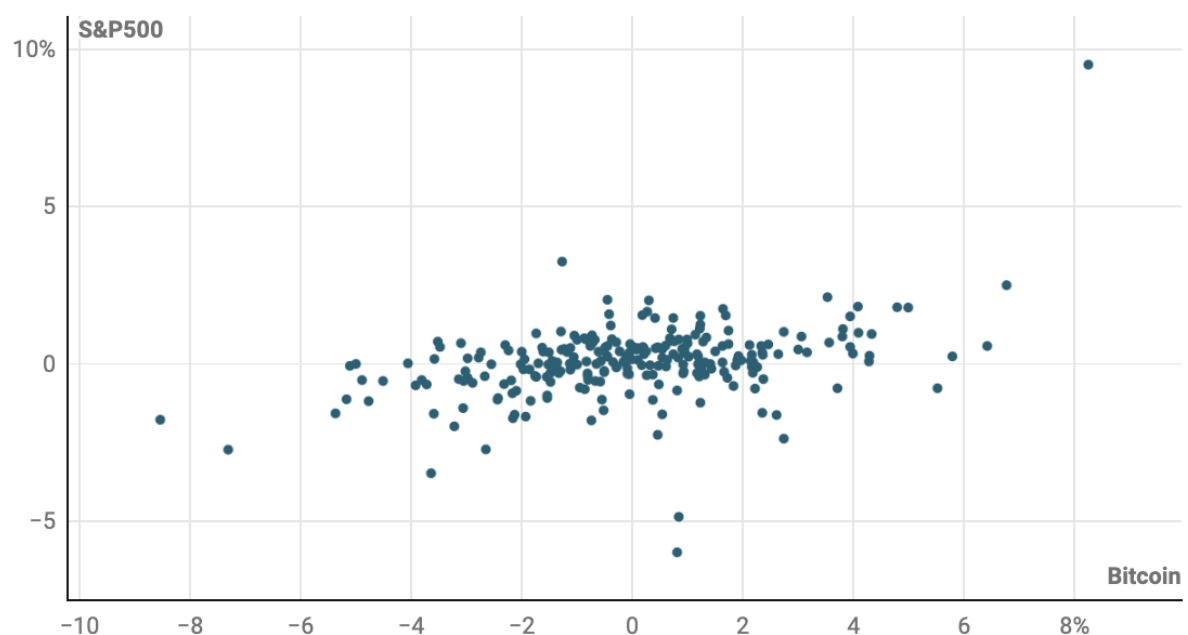


(Click on the link in the image to view the chart in more detail).

 **Conclusion on risks:** Volatility and drawdown charts confirm that the S&P 500 remained a relatively stable asset in 2025, with moderate fluctuations and predictable growth. In contrast, Bitcoin turned out to be a ship in a storm: its extreme volatility and deep drawdowns brought investors pain rather than profit. This underscores its role not as “digital gold,” but as a speculative asset with a risk 3–4 times higher than that of the stock market.

4. Correlation between assets and the macroeconomic impact.

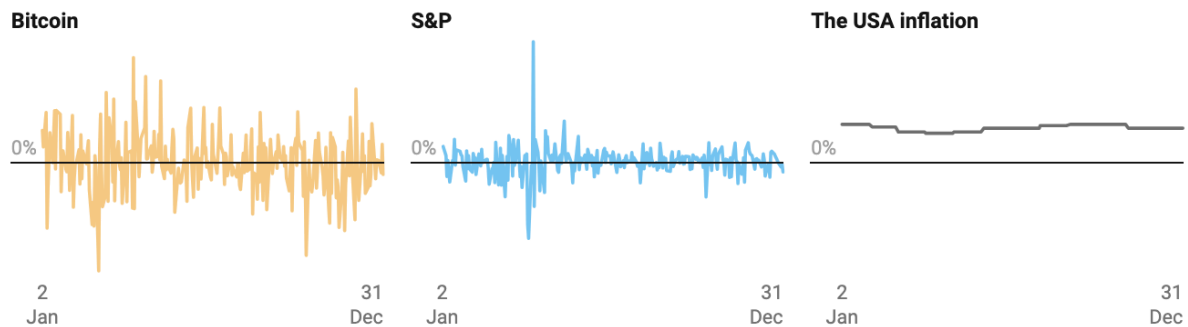
Assets correlation



The estimated Pearson correlation coefficient between Bitcoin and S&P 500 returns for 2025 was **0.4**, indicating a moderate positive correlation between the two assets.

 **The main point:** For most of the year, asset prices moved in a relatively consistent direction, and the correlation would have been much higher, but everything changed in October 2025, when cryptocurrency prices plummeted sharply due to billions in liquidations and market overheating.

The Impact of Microeconomics



*The inflation rate data for October is based on September figures because, due to the government shutdown, the rates were not announced.

 **The main point: The index shows high sensitivity to macroeconomic events (in particular, a sharp reaction to the imposition and removal of tariffs in April 2025). Bitcoin maintains high internal volatility regardless of the geopolitical agenda, exhibiting dynamics driven by its own market factors rather than external shocks. Of course, macroeconomics does influence Bitcoin, but not as much as traditional markets.**

Conclusion of the comparative analysis of Bitcoin vs. the S&P 500:

There are two main types of investment portfolios: aggressive and conservative.

Bitcoin: In my opinion, it is not suitable for either portfolio, as the risks are too high and there is no reward for them. Moreover, an investor will pay for this asset in their portfolio not only with a negative balance but also with their nerves and mental well-being. This asset follows its own path, not mirroring the movements of other assets. Unlike the stock market and other markets, Bitcoin is slightly influenced by external factors, making it unpredictable. Its movements on the chart resemble a ship in a storm: you don't know where it will end up in 5 minutes.

Index: the complete opposite. This is a benchmark index with a solid reputation, so it's a common holding in many people's portfolios. It suits both types of portfolios: risk hedging in an aggressive strategy and capital accumulation in a conservative one. This asset has predictable movements, low volatility, and pays dividends. But the S&P also has a weakness: the macroeconomy. Negative news can create panic in the market, which can lead to a sharp decline.

